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$$\int \frac{x^3+1}{x-1}dx = \int \frac{x^3-1+2}{x-1}dx = \int \frac{(x-1)(x^2+x+1)}{x-1}dx + \int \frac{2}{x-1}dx$$
$$= \int (x^2+x+1)dx + \int \frac{2}{x-1}dx = \int x^2dx + \int xdx + \int 1dx + \int \frac{2}{x-1}dx$$

$u = x-1 \Rightarrow du = dx$

$$\int \frac{2}{x-1}dx = \int \frac{2}{u}du = 2\ln|u| + c = 2\ln|x-1| + c$$
$$\int \frac{x^3+1}{x-1}dx = \frac{x^3}{3} + \frac{x^2}{2} + x + 2\ln|x-1| + c$$

References